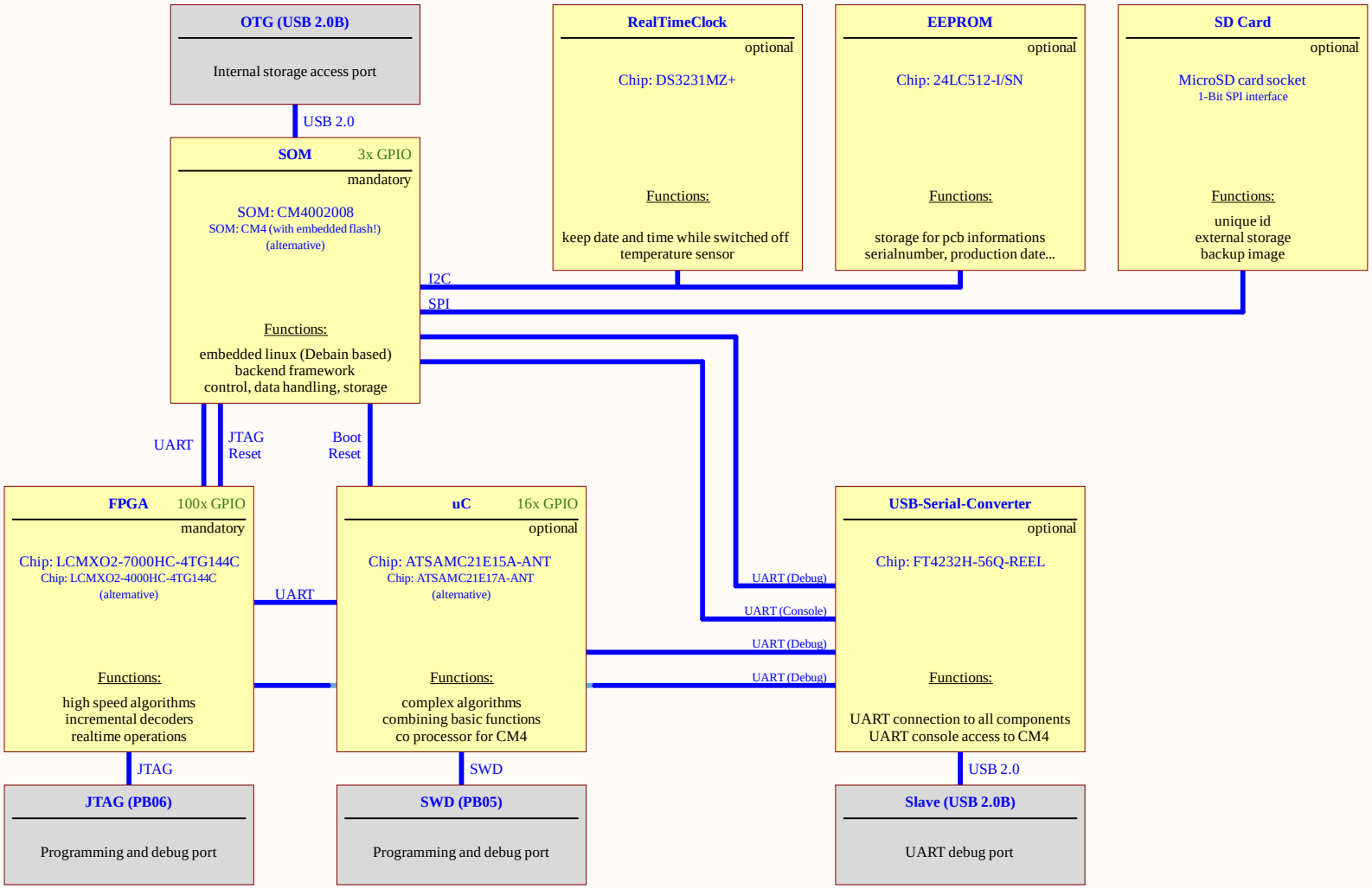
KOM ACB23 4009

Overview Control Unit for ACB23



Title	*Overview*		
Size: A4	Number: 2023-03-28	Revision: v0.11	
Date: 14.08.2023	Time: 11:30:29	Sheet 1 of 9	
File: 01_KOM_ACB23_4009_Overview.SchDoc			

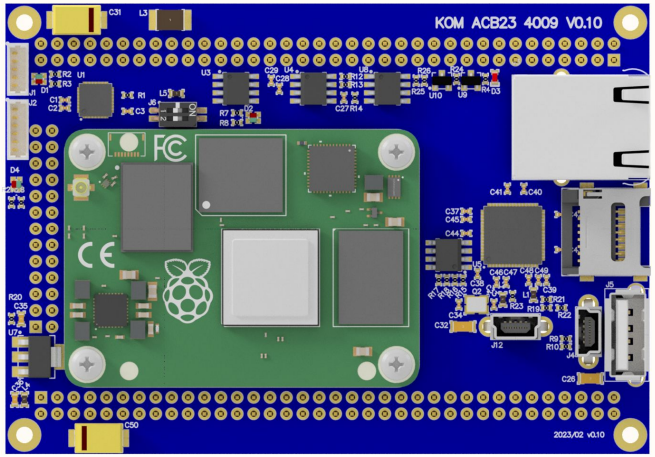
Confidential

Vietzke Engineering
Rosengasse 21
98693 Ilmenau

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KOM ACB23 4009

Overview PCB



SWD
ATSAMC21

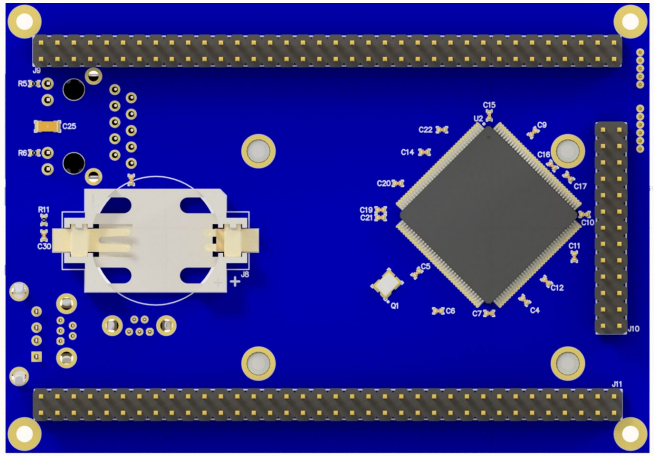
JTAG
MachXO2

Gigabit ethernet port

Cannot open file D:\Dropbox\Arbeit\Kunden\Kompass\Projects\KOM_ACB23_4009\Hardware\Documents\2023-03-28 KOM_ACB23_4009 FrontView.png. File does not exist.

MicroSD slot

J5: USB Host / J4: Slave of CM4
J12: FTDI USB Connector



CR2032
RTC Backup

Title	***Overview PCB***		
Size: A4	Number: 2023-03-28	Revision: v0.11	
Date: 14.08.2023	Time: 11:30:29	Sheet 2 of 9	
File: 02_KOM_ACB23_4009_Overview_PCB.SchDoc			

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KOM ACB23 4009

Remarks, History and Changelog

Fault list

Version 0.10

- R27 / R28 Designator inside component!
- C42 / C43 collision SD Card!
- J4 to close to USB A
- Add addional components

Changelog

Version 0.10
- first version

Version 0.11
- R27 / R28 designators moved
- C42 / C43 moved away from SD socket
- Added 1mm extra space between USB sockets
- Added additional components

Title	***Remarks***			***Confidential***
Size: A4	Number: 2023-03-28	Revision: v0.11		***Vietzke Engineering*** *Rosengasse 21 98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:29	Sheet 3 of 9		
File: 03_KOM_ACB23_4009_Remarks.SchDoc				

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KOM ACB23 4009

Additional components (Mechnical, Screws, Cable etc.)

Mechanical

M2.5 9774015151R

M2.5 9774015151R

M2.5 9774015151R

M2.5 9774015151R

DIN912 M2.5x4

DIN912 M2.5x4

DIN912 M2.5x4

DIN912 M2.5x4

Misc



CR2032



MicroSD Card

Cables

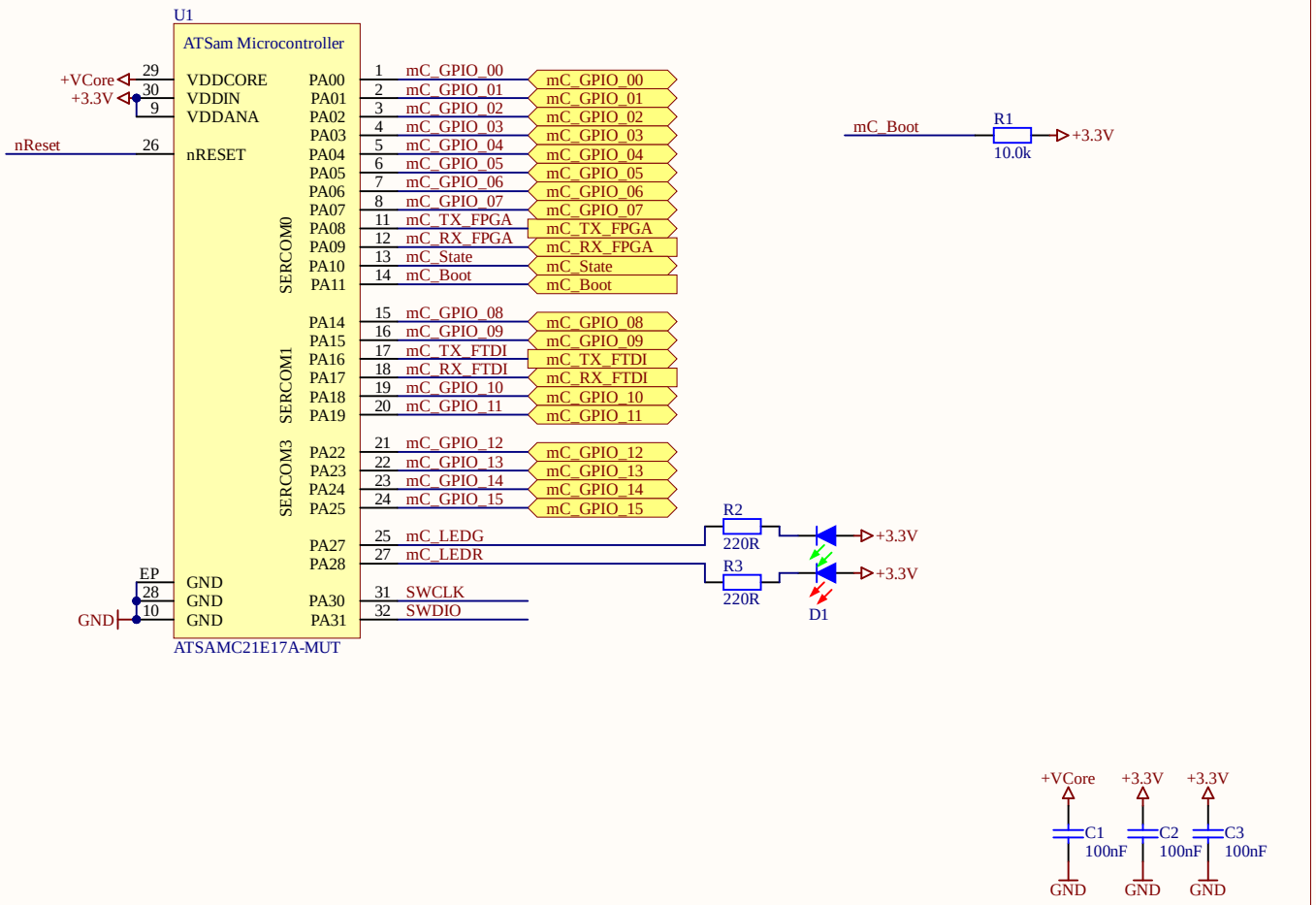
Title *Addtional components*			*Confidential*
Size: A4	Number: 2023-03-28	Revision: v0.11	*Vietzke Engineering* *Rosengasse 21* *98693 Ilmenau*
Date: 14.08.2023 Time: 11:30:30		Sheet 4 of 9	
File: 04_KOM_ACB23_4009_AdditionalComponents.SchDoc			

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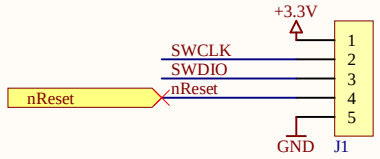
KOM ACB23 4009

Microcontroller with programming port.

Microcontroller



Programming connector



Title **MicroController**			*Confidential*
Size: A4	Number: 2023-03-28	Revision: v0.11	*Vietzke Engineering* *Rosengasse 21* *98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:30	Sheet 5 of 9	Cannot open file D:\Dropbox\Arbeit\Altium
File: 05_KOM_ACB23_4009_MicroController.SchDoc			

KOM ACB23 4009

Central FPGA for real time control.

FPGA

U2E
FPGA
PL10A 13 GPIO_Top_29
PL10B 14 GPIO_Top_13
PL11A 15 GPIO_Top_30
PL11B 17 GPIO_Top_14
PL12A, PCLKT4_0 19 GPIO_Top_31
PL12B, PCLKC4_0 20 GPIO_Top_15
PL15A 21 GPIO_Top_16
PL15B 22 GPIO_Top_00
PL17A 23 GPIO_Top_17
PL17B 24 GPIO_Top_01
LCMXO2-7000HC-4TG144C

U2D
FPGA
PL19A 25 GPIO_Top_18
PL19B 26 GPIO_Top_02
PL22A, PCLKT3_0 27 GPIO_Top_19
PL22B, PCLKC3_0 28 GPIO_Top_03
PL23D 31 GPIO_Top_20
PL24A 32 GPIO_Top_04
PL24B 33 GPIO_Top_21
PL25A 34 GPIO_Top_05
PL25B 35 GPIO_Top_22
LCMXO2-7000HC-4TG144C

U2F
FPGA
PL3A, L_GPLLT_FB 1 GPIO_Top_24
PL3B, L_GPLLC_FB 2 GPIO_Top_08
PL4A, L_GPLLT_IN 3 GPIO_Top_25
PL4B, L_GPLLC_IN 4 GPIO_Top_09
PL6A, PCLKT5_0 5 GPIO_Top_26
PL6B, PCLKC5_0 6 GPIO_Top_10
PL8A 9 GPIO_Top_27
PL8B 10 GPIO_Top_11
PL9A 11 GPIO_Top_28
PL9B 12 GPIO_Top_12
LCMXO2-7000HC-4TG144C

Q1
Crystal Oscillator
+3.3V 1 Enable
+3.3V 4 VCC
GND 2 GND
OUT 3 ClockFPGA
48.0 Mhz

+3.3V
1 FPGA_TDO FPGA_TDO
2 FPGA_TDI FPGA_TDI
3 FPGA_TCK FPGA_TCK
4 FPGA_TMS FPGA_TMS
5
6 GND
J2

FPGA_LED_Red R27 220R +3.3V
FPGA_LED_Green R28 220R D4 +3.3V

U2G
FPGA
VCC 144 +3.3V
VCC 108
VCC 72
VCC 36
GND 134 GND
GND 24 VCCIO0 118
GND 16 VCCIO0 123
GND 101 VCCIO0 133
GND 90 VCCIO1 79
GND 80 VCCIO1 88
GND 8 VCCIO1 102
GND 18 VCCIO2 37
GND 29 VCCIO2 51
GND 46 VCCIO2 66
GND 53 VCCIO3 30
GND 64 VCCIO4 16
GND VCCIO5 7
LCMXO2-7000HC-4TG144C

+3.3V
C4 100nF C5 100nF C6 100nF C7 100nF
GND
+3.3V
C9 100nF C10 100nF C11 100nF C12 100nF
GND
+3.3V
C14 100nF C15 100nF C16 100nF C17 100nF
GND
+3.3V
C19 100nF C20 100nF C21 100nF C22 100nF
GND

U2C
FPGA
PB4A 38 GPIO_Top_06
PB4B 39 GPIO_Top_23
PB6A, CSSPIN/MD4/TDOB 40 GPIO_Top_07
PB6B 41 GPIO_Bottom_28
PB9A 42 GPIO_Bottom_57
PB9B 43 GPIO_Bottom_27
PB12A, MCLK/CCLK 44 GPIO_Bottom_56
PB12B, SO/SPISO/IO1/MD1/TDIL 45 GPIO_Bottom_26
PB13A 47 GPIO_Bottom_55
PB13B 48 GPIO_Bottom_25
PB16A, PCLKT2_0/INTEST_OVER 49 ClockFPGA
PB16B, PCLKC2_0 50 GPIO_Bottom_54
PB18A 52 GPIO_Bottom_24
PB18B 54 GPIO_Bottom_53
PB23A, PCLKT2_1/INTEST_OVER 55 nReset
PB23B, PCLKC2_1 56 GPIO_Bottom_23
PB26A 57 GPIO_Bottom_52
PB26B 58 GPIO_Bottom_22
PB29A 59 GPIO_Bottom_51
PB29B 60 GPIO_Bottom_21
PB31A 61 GPIO_Bottom_50
PB31B 62 GPIO_Bottom_20
PB31D 63 GPIO_Bottom_49
PB35A 65 GPIO_Bottom_19
PB35B 67 GPIO_Bottom_48
PB37A 68 GPIO_Bottom_18
PB37B 69 GPIO_Bottom_47
PB38A, SN/MD5/SCAN_SHFT_ENB/TDIB 70 GPIO_Bottom_17
PB38B, SI/SISPI/IO0/MD0/TDOR 71 GPIO_Bottom_46
LCMXO2-7000HC-4TG144C

U2A
FPGA
PT9A 143 FPGA_RX_FTDI
PT9B 142 FPGA_TX_FTDI
PT10A 141 mC_TX_FPGA
PT10B 140 mC_RX_FPGA
PT11A 139 CM_TX_FPGA
PT11B 138 CM_RX_FPGA
PT14C, TDO 137 FPGA_TDO
PT14D, TDI/MD7 136 FPGA_TDI
PT15A 135 FPGA_State
PT15B 132 FPGA_LED_Red
PT17C, TCK/TEST_CLK 131 FPGA_TCK
PT17D, TMS 130 FPGA_TMS
PT18A, PCLKT0_1 128 FPGA_LED_Green
PT18B, PCLKC0_1 127 GPIO_Left_08
PT22C, SCL/IO2/PCLKT0_0 126 GPIO_Left_09
PT22D, SDA/IO3//PCLKC0_0/TDOT 125 GPIO_Left_06
PT25A 122 GPIO_Left_07
PT25B 121 GPIO_Left_04
PT27C, JTAGENB/MD6/TDIR 120 GPIO_Left_05
PT27D, PROGRAMN 119 GPIO_Left_02
PT28A 117 GPIO_Left_03
PT28B 115 GPIO_Left_00
PT33A 114 GPIO_Left_01
PT33B 113 GPIO_Bottom_29
PT35A 112 GPIO_Bottom_00
PT35B 111 GPIO_Bottom_30
PT36C, INITN 110 GPIO_Bottom_01
PT36D, DONE 109 GPIO_Bottom_31
LCMXO2-7000HC-4TG144C

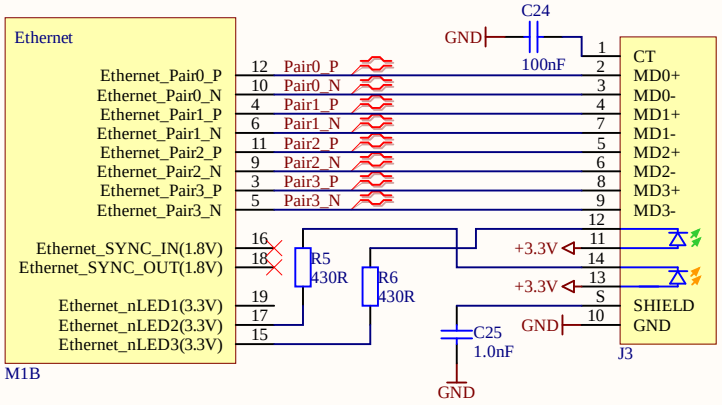
U2B
FPGA
PR2A, R_GPLLT_FB (DQ0) 107 GPIO_Bottom_02
PR2B, R_GPLLC_FB (DQ0) 106 GPIO_Bottom_32
PR3A, R_GPLLT_IN (DQ0) 105 GPIO_Bottom_03
PR3B, R_GPLLC_IN (DQ0) 104 GPIO_Bottom_33
PR4C 103 GPIO_Bottom_04
PR5A (DQ0) 100 GPIO_Bottom_34
PR5B (DQ0) 99 GPIO_Bottom_05
PR7A (DQ0) 98 GPIO_Bottom_35
PR7B (DQ0) 97 GPIO_Bottom_06
PR9A (DQ0) 96 GPIO_Bottom_36
PR9B (DQ0) 95 GPIO_Bottom_07
PR11A (DQS0) 94 GPIO_Bottom_37
PR11B (DQS0N) 93 GPIO_Bottom_08
PR12A, PCLKT1_0 (DQ0) 92 GPIO_Bottom_38
PR12B, PCLKC1_0 (DQ0) 91 GPIO_Bottom_09
PR15A 89 GPIO_Bottom_39
PR15B 87 GPIO_Bottom_10
PR16A (DQS1) 86 GPIO_Bottom_40
PR16B (DQS1N) 85 GPIO_Bottom_11
PR17A (DQ1) 84 GPIO_Bottom_41
PR17B (DQ1) 83 GPIO_Bottom_12
PR18A (DQ1) 82 GPIO_Bottom_42
PR18B (DQ1) 81 GPIO_Bottom_13
PR21A (DQ1) 78 GPIO_Bottom_43
PR21B (DQ1) 77 GPIO_Bottom_14
PR23A (DQ1) 76 GPIO_Bottom_44
PR23B (DQ1) 75 GPIO_Bottom_15
PR24A (DQ1) 74 GPIO_Bottom_45
PR24B (DQ1) 73 GPIO_Bottom_16
LCMXO2-7000HC-4TG144C

Title **FPGA**			**Confidential**
Size: A4	Number: 2023-03-28	Revision: v0.11	*Vietzke Engineering* *Rosengasse 21* *98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:30	Sheet 6 of 9	Cannot open file D:\Dropbox\Arbeit\Altium
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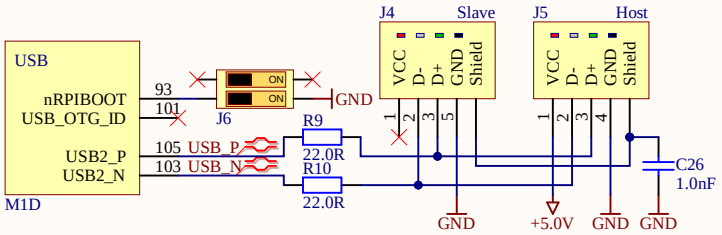
KOM ACB23 4009

Compute module 4 with its peripherals.

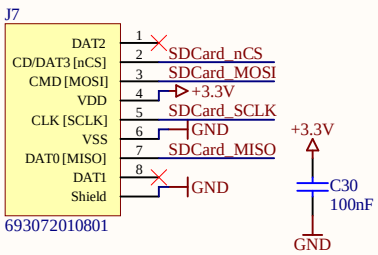
Gigabit Ethernet



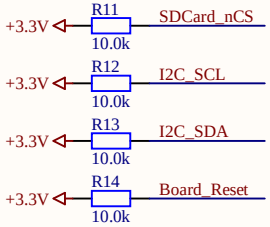
USB Slave / Host



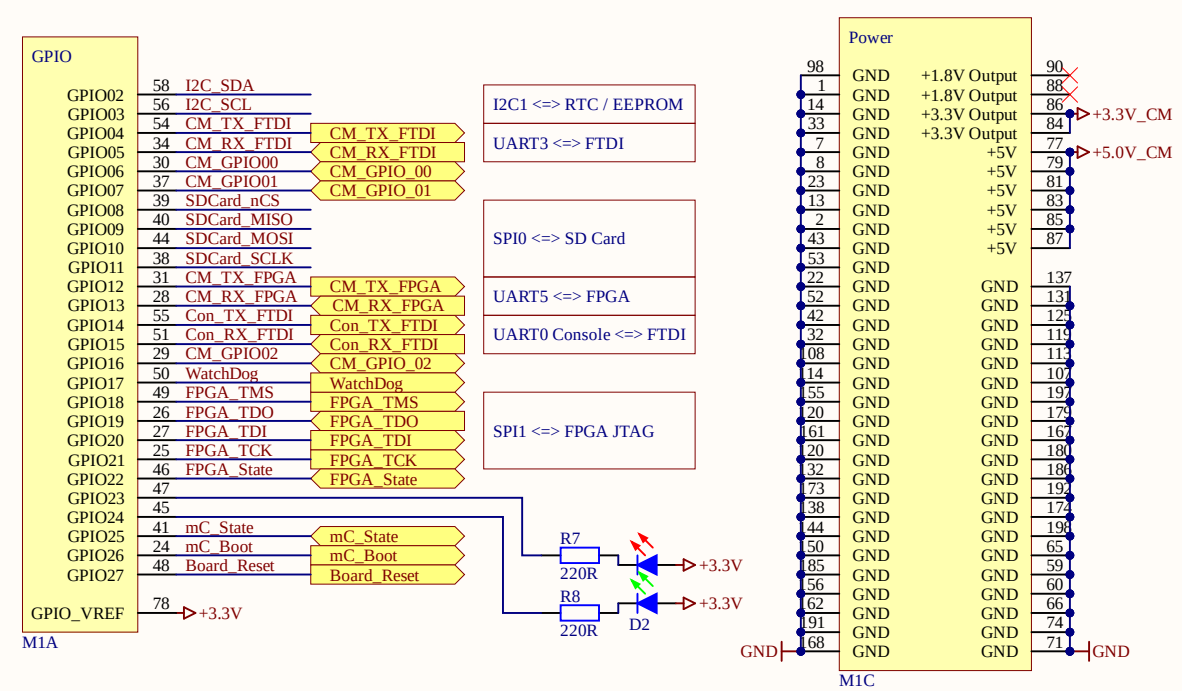
MicroSD Card (SPI Mode, no license fee)



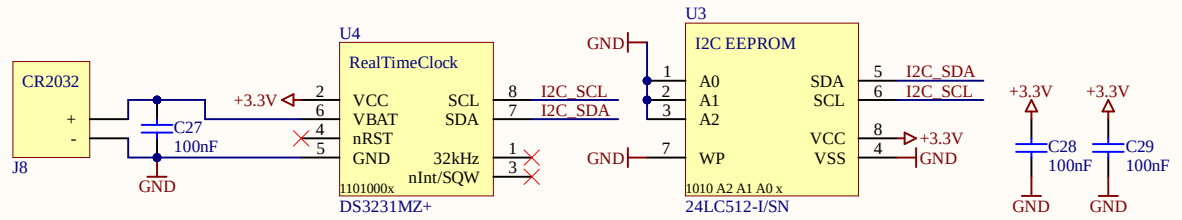
Pull resistors



Compute Module 4



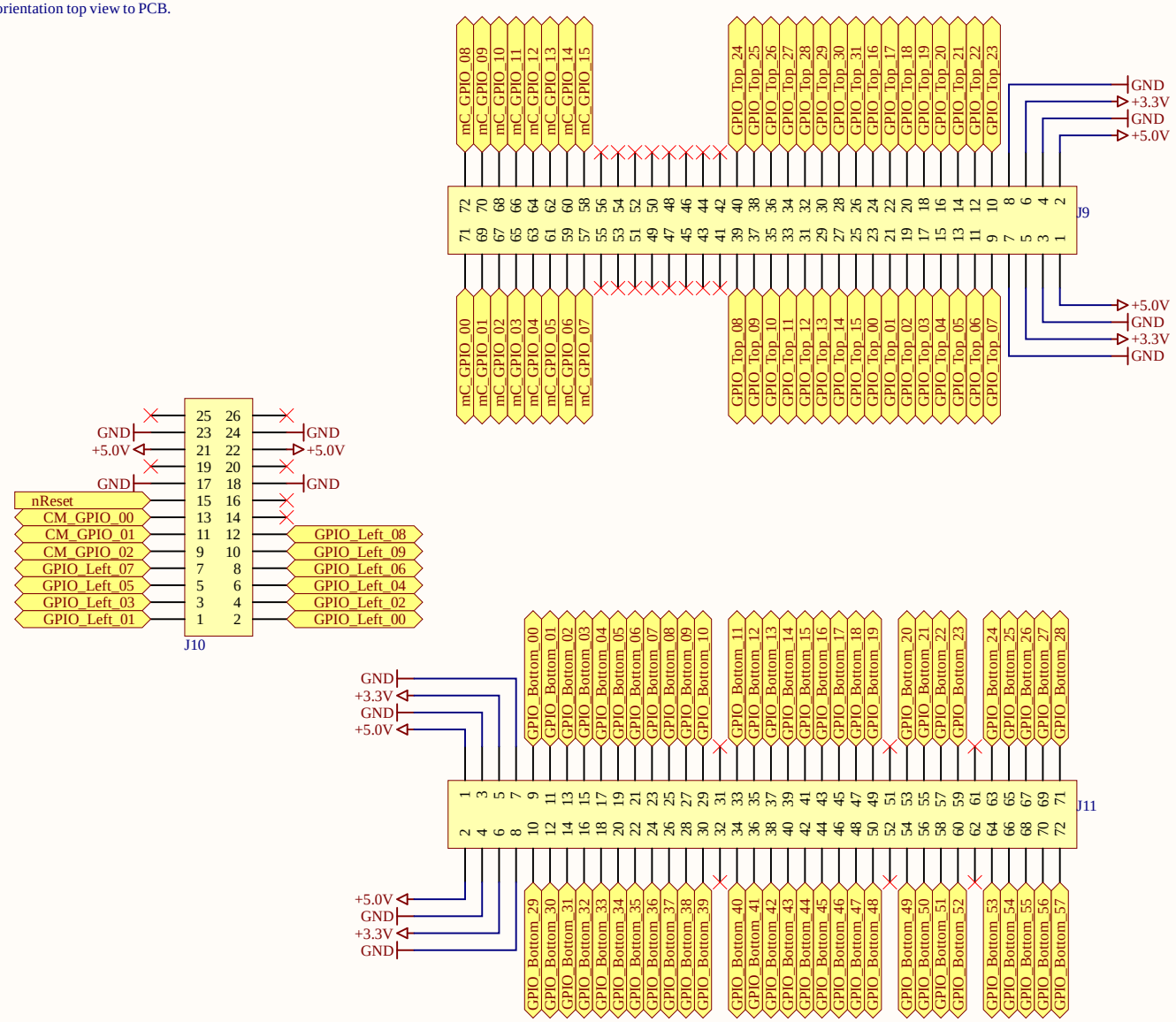
I2C RealTimeClock and EEPROM



Title **ComputeModule**			***Confidential***
Size: A4	Number:2023-03-28	Revision:v0.11	***Vietzke Engineering*** *Rosengasse 21 98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:30	Sheet 7 of 9	Cannot open file D:\Dropbox\Arbeit\Altium
File: 07_KOM_ACB23_4009_ComputeModule.SchDoc			

KOM ACB23 4009

Baseboard connectors. Component orientation top view to PCB.

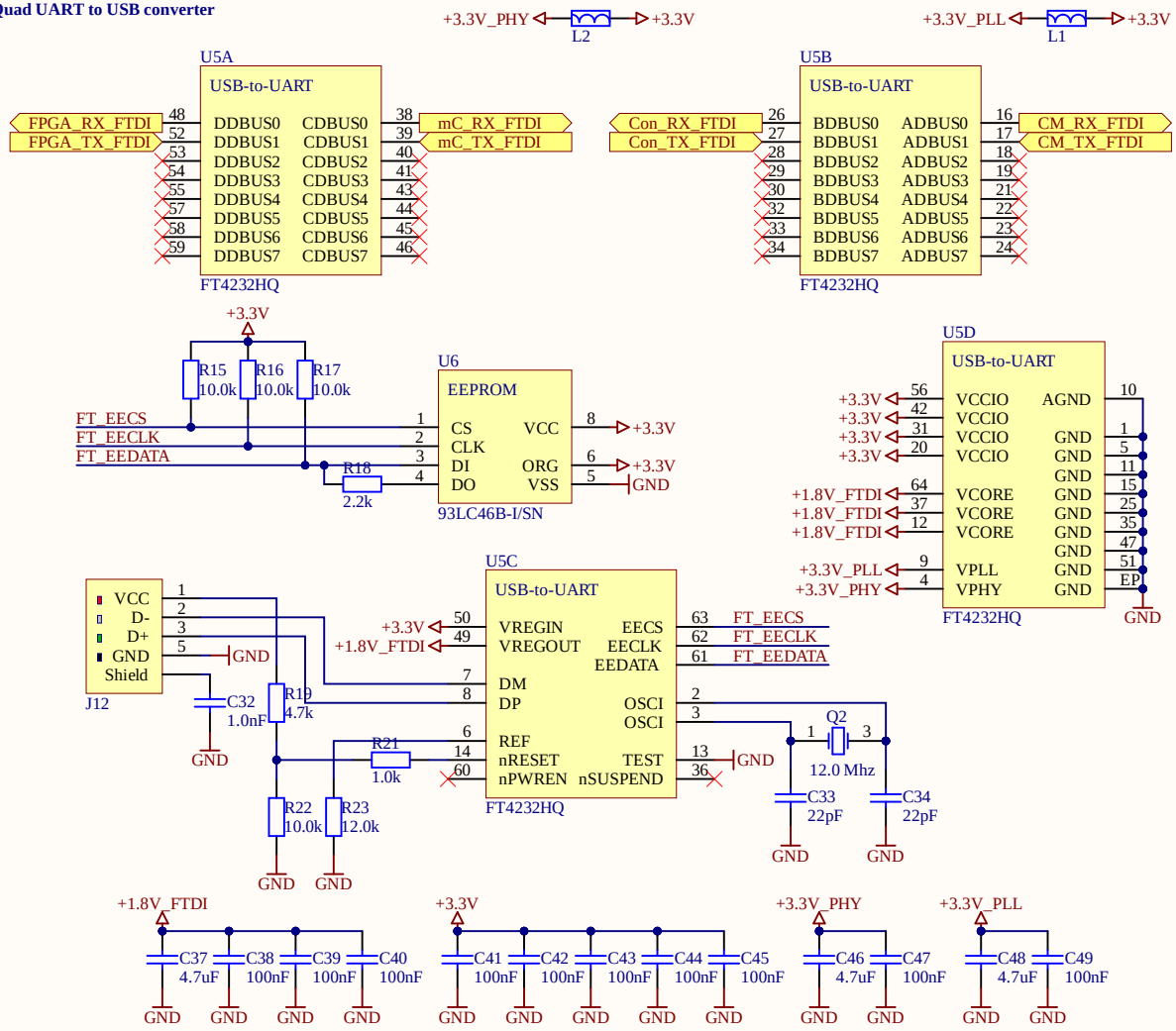


Title **Connectors**			*Confidential*
Size: A4	Number: 2023-03-28	Revision: v0.11	*Vietzke Engineering* *Rosengasse 21* *98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:30	Sheet 8 of 9	Cannot open file D:\Dropbox\Arbeit\Altium
File: 08_KOM_ACB23_4009_Connectors.SchDoc			

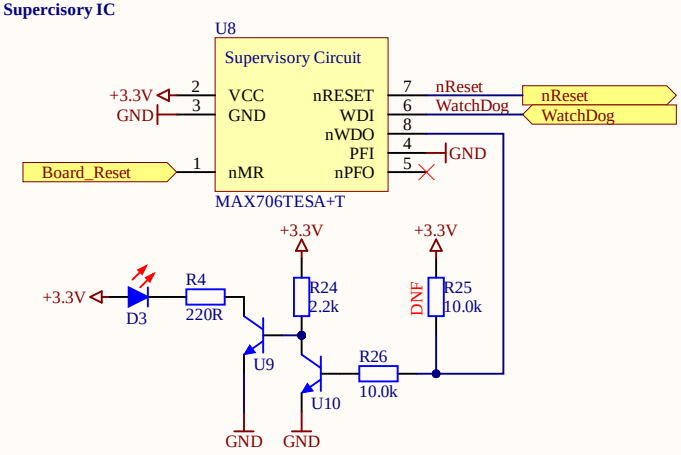
KOM ACB23 4009

Power supply and debug FTDI usb port.

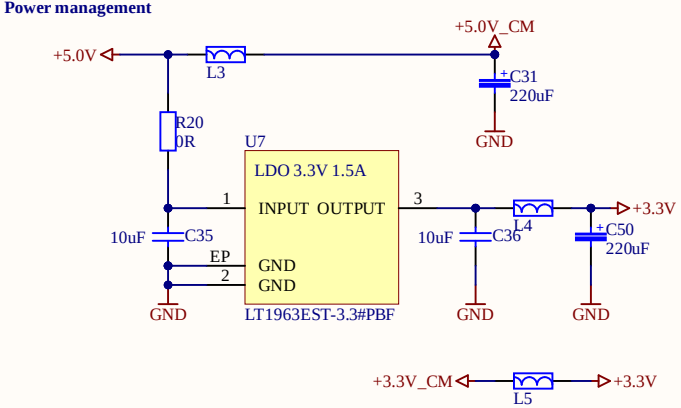
Quad UART to USB converter



Supercisory IC



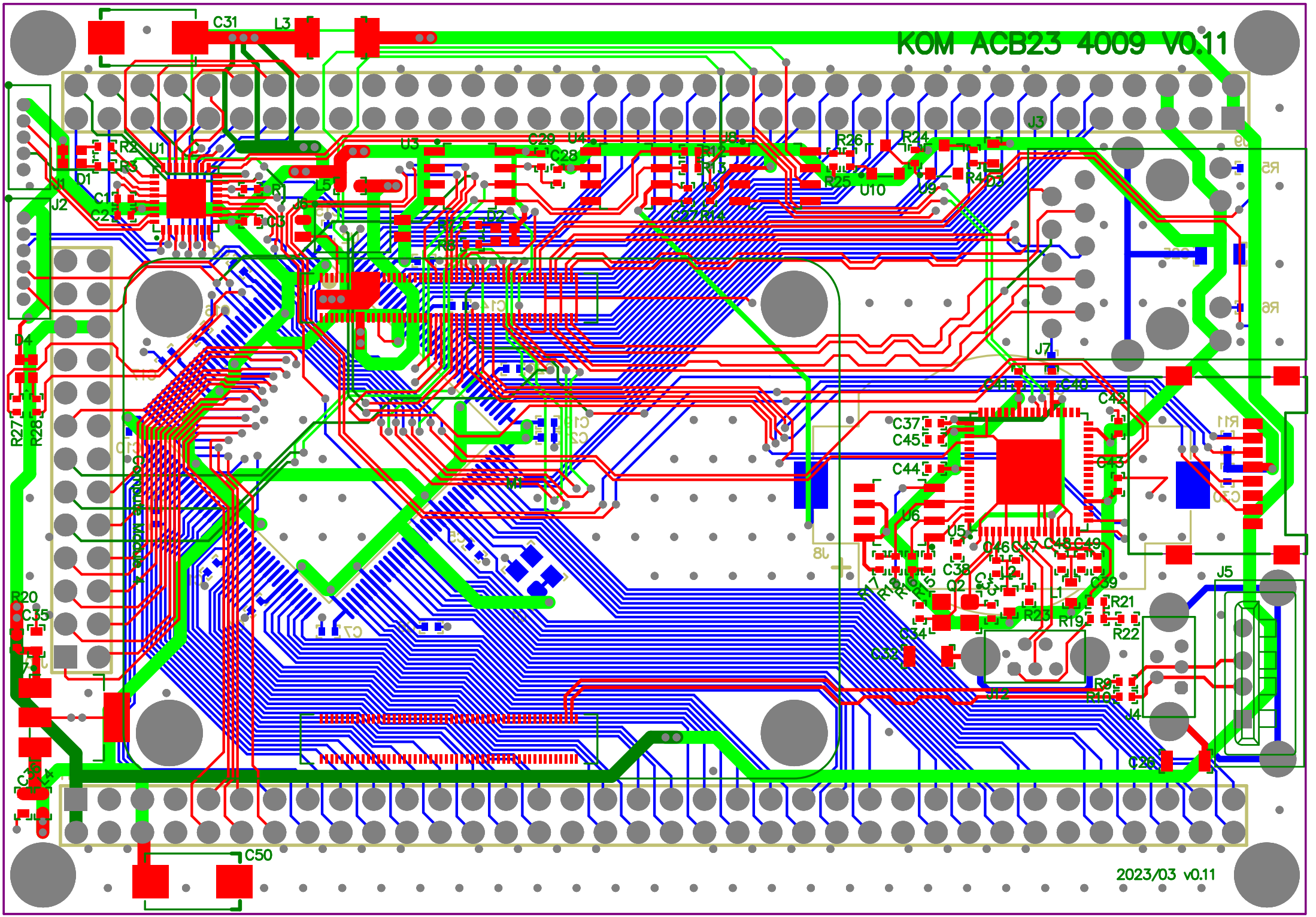
Power management



Option A: add R20, L4. do not fit L5
Use LDO U7 to generate 3.3V (up to 1.5 amps)

Option B: add L5, do not fit R20, C35, U7, C36, L4
Use internal 3.3V generation of CM4 (up to 0.6 amps)

Title	***Periphery***		***Confidential***
Size: A4	Number: 2023-03-28	Revision: v0.11	***Vietzke Engineering*** *Rosengasse 21 98693 Ilmenau*
Date: 14.08.2023	Time: 11:30:30	Sheet 9 of 9	Cannot open file D:\Dropbox\Arbeit\Altium
File: 09_KOM_ACB23_4009_Periphery.SchDoc			


KOM ACB23 4009 V0.11
2023/03 v0.11